In nephrotic syndrome, heavy protein loss in the urine lowers plasma oncotic (colloid osmotic) pressure, and that fall drives the oedema. Note that "plasma osmotic pressure" in this context means oncotic pressure, the part of osmotic pressure due to plasma proteins, mainly albumin.

1. The sequence

1. **Glomerular injury:** damage to the glomerular filtration barrier (podocyte injury and loss of the negatively charged basement membrane) makes it leaky to protein. Causes include minimal change disease, membranous nephropathy, FSGS, and diabetic nephropathy.
2. **Massive proteinuria:** more than 3.5 g/day in adults. Albumin, being small and negatively charged, is lost most.
3. **Hypoalbuminaemia:** the liver cannot synthesize albumin fast enough to replace the loss, so serum albumin falls (below about 30 g/L).
4. **Fall in plasma oncotic pressure:** albumin generates about 75-80% of plasma oncotic pressure, so this drops sharply.
5. **Starling forces shift:** oncotic pressure normally pulls fluid back into capillaries at the venous end. When it falls, hydrostatic pressure now outweighs it, and fluid moves out into the interstitium.
6. **Oedema:** typically periorbital (morning), then legs, sacrum, scrotum or labia, and ascites or pleural effusions in severe cases.

2. Why the oedema persists: renal salt and water retention

- Fluid leaving the vessels reduces effective circulating volume.
- This activates **RAAS**, ADH, and the sympathetic system, so the kidneys retain sodium and water.
- This retained fluid also leaks out because oncotic pressure stays low, so oedema worsens.

The "underfill" model above (low oncotic pressure causing volume depletion) fits children with minimal change disease. In many adults, an "overfill" mechanism also operates: primary sodium retention by the kidney (e.g., epithelial sodium channel activation in the collecting duct) occurs independently of low circulating volume. This is why some patients have hypertension and a normal or high blood volume despite hypoalbuminaemia.

3. Related features of the syndrome

- **Hyperlipidaemia:** low oncotic pressure stimulates hepatic synthesis of lipoproteins, and lipoprotein lipase activity falls. Lipiduria produces fatty casts and "oval fat bodies."
- **Hypercoagulability:** loss of antithrombin III and other anticoagulant proteins in urine, plus increased hepatic synthesis of fibrinogen, raises the risk of renal vein thrombosis and DVT.
- **Infection risk:** loss of immunoglobulins and complement factors.

4. Comparison with liver disease

Both cause hypoalbuminaemia and oedema. In nephrotic syndrome, albumin is **lost** through the kidney. In cirrhosis, albumin is **under-produced**, and portal hypertension is the main driver of ascites.

Summary: glomerular damage → proteinuria → hypoalbuminaemia → low plasma oncotic pressure → fluid shift into the interstitium → oedema, amplified by secondary sodium and water retention.